

27	C	The range of wavelengths for visible light is 400 nm to 700 nm. Since $E = \frac{hc}{\lambda}$, the energies of these photons range from $\frac{(6.63 \times 10^{-34})(3.00 \times 10^8)}{(700 \times 10^{-9})(1.60 \times 10^{-19})} = 1.7759 \text{ eV to } \frac{(6.63 \times 10^{-34})(3.00 \times 10^8)}{(400 \times 10^{-9})(1.60 \times 10^{-19})} = 3.1078 \text{ eV .}$ Only 3 transitions will result in emissions of such photons: 6.12 – 4.28 = 1.84 eV 6.81 – 4.28 = 2.53 eV 7.02 – 4.28 = 2.74 eV
28	D	For the cut off wavelength